Indian Institute of Technology, Bhilai

CSP203: Software Tools & Technologies Laboratory

Lab Activity - 1

Full Marks: 50 *Deadline: 11:59 PM, 02-08-2025*

Submission Instructions:

1) Answer all the questions.
2) Execute all the commands in linux and take the screenshots.
3) Prepare a pdf which contains the question number and corresponding screenshots of the outputs.
4) Naming of the file should strictly follow the given format:
   <Roll_No>_<Name>_<Activity_No>
   If your name is Shareen, roll number is B24CS055 then the filename should be:
   B24CS055_Shareen_01.pdf
5) **Plagiarism is strictly prohibited.**

Objectives:

1) Getting familiar with the linux operating system.
2) Introduction to the file system and directory structure in linux.
3) Seeing overview of different file system management commands and directory management commands.
4) Getting hands-on experience of administrative privileges of linux systems and changing access permission of files, directories and users.
5) Leaning execution of process management commands and usage of the commands for manipulation of the processes.

Task 1: Overview of all concepts [10 Marks]

You have to create a directory named <Your_Roll_No>_<Name>. Now enter into the directory. Create Five more directories dir1, dir2, dir3, dir4, dir5. Enter into the dir1. Create the files file1.txt, file2.txt, file3.txt, file4.txt, file5.py. [2 marks]

Now execute the following instructions:

1. Now you are in dir1. Don't move to any other directory. Copy the file1.txt to dir4 and then rename the file as file6.txt. Change the file permissions as follows:
    a. User: Read and Write
    b. Group: Read and Execute
    c. Others: Only read [3 (1+1+1) Marks]
2. Now move to dir5. Staying in dir5 move file3.txt to this directory. Change the file 3 permission as follows:
    a. User: Read, write and execute
    b. Group: Read, write and execute
    c. Others: Read, write and execute [1 marks]
3. Make a copy of file3.txt as copy_file3.txt. Delete write permission from group and user. [Don't use chmod 755]. [1 marks]
4. Make a user named <your_name>_02 and set the password as "password". Add the user to the sudo group. Log in to the root user of the linux and go to the home directory of <your_name>_02. Now copy the directory <Your_Roll_No>_<Name> to the home directory of the <your_name>_02 user. [2 marks]
5. Remove <your_name>_02 user from the sudo group. [1 marks]
6. Now logout from the root user and delete <your_name>_02 user completely including any mail spool, home directory or files. [1 marks]
7. Go to the dir3 using one command only. Copy file6.txt to the current directory. Rename the file as file7.txt. Now go outside of <Your_Roll_No>_<Name> directory. Delete the whole directory using one command only. [1 marks]

[Instruction: After every step you have to show the changes using listing command and take screenshots]

Task 2: Date and process management [14 Marks]

Now you have to manipulate process execution by following the given commands:

1. List all the running processes of the system. [1 marks]
2. Open the firefox. Now in the terminal find the process ids for firefox. [2 marks]
3. Kill all the processes containing process ids of firefox with SIGKILL signal. [1 marks]
4. Write a c program to print hello world. Now measure the command runtime of the c program using linux command. [2 marks]
5. Show all the processes for your user. [1 marks]
6. Print the calendar of 2050 in julian day format. [1 marks]
7. Show the current time and date in the following format: [2 marks]

```
Time: 12:32:41 PM
Date: 28-07-2025
Month Name: July
Weekday: Monday
Quarter of the year: 3
Time Zone: +05:30:00
```

8. Display the universal date by 2 years 5 months 11 days 3 hours 22 minutes 20 seconds ago. [2 marks]
9. Print the date and time 02:50:26 PM of 12th March 2027 in the following format: (0: Sunday …. 6: Saturday and 24 hours format) [2 marks]

```
13:22:23
Date: 02/02/26
Month Name: February
Weekday: Monday(Weekday Number: 1)
Quarter of the year: 1
Time Zone: +05:30:00
```

Task 3: Directory, file management and introduction to text processing

[10 Marks]

1. Create a folder on your Desktop named with your Roll Number (e.g., CS24XXX). [1 marks]
2. Inside that folder, create a new sub-folder named FileOps/. [1 marks]
3. In the FileOps/ directory, create three files:
   report.txt

data.txt
confidential.txt [1 marks]

Add sample content using echo in each file.

4. Perform the following operations:
   A. Copy report.txt to a new file named report_backup.txt.
   B. Rename data.txt to dataset.csv.
   C. Move confidential.txt to a new folder Secure/ inside FileOps/.
   D. Rename the FileOps as SystemFiles [2 marks]
5. Set the following permissions using chmod:
   report.txt: Read and write for the owner only.
   report_backup.txt: Read-only for all users.
   dataset.csv: Execute permission for all users.
   confidential.txt (inside Secure/): No permissions for others (only the owner can read/write). [3 marks]
6. Display and save permissions in a file permissions.txt [2 marks]

[Instruction: After every step you have to show the changes using listing command and take screenshots]

Task 4: General user commands[10 marks]

1. Display all currently running processes for the current user [1 marks]
2. Identify and display the PID and CPU usage of the top 5 memory-consuming processes [1 marks]
3. Print the system hostname and OS name [1 marks]
4. Show the current system date and time [1 marks]
5. Print all environment variables and store them in env.txt. [1 marks]
6. Display all currently running processes for the current user [1 marks]
7. Identify and display the PID and CPU usage of the top 5 memory-consuming processes [1 marks]
8. Print the system hostname and OS name [1 marks]
9. Show the current system date and time [1 marks]
10. Print all environment variables and store them in env.txt. [1 marks]

[Instruction: After every step you have to show the changes using listing command and take screenshots]

Task 5: Text processing [6 Makrs]

1. Use echo to create a file named greetings.txt with the line:
   "Welcome to Linux Lab" [1 marks]
2. Append two more lines to greetings.txt using echo. [1 marks]
3. Display the contents of the file. [1 marks]
4. Use more and less to view the contents page-by-page (add 20 more lines using a loop or copy-paste to make the file longer). [1 marks]
5. Display the first 5 lines of the file. [1 marks]
6. Display the last 15 lines of the file. [1 marks]